

Sample Project Documentation

ComicCraft - AI Comic Story Creator using Gemini Models

Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester

Project period: 1 Oct 2026 - 2 Oct 2026

1 Introduction

ComicCraft is a web app that turns a short story idea into a five-panel illustrated comic and PDF using Google Gemini and Stable Diffusion.

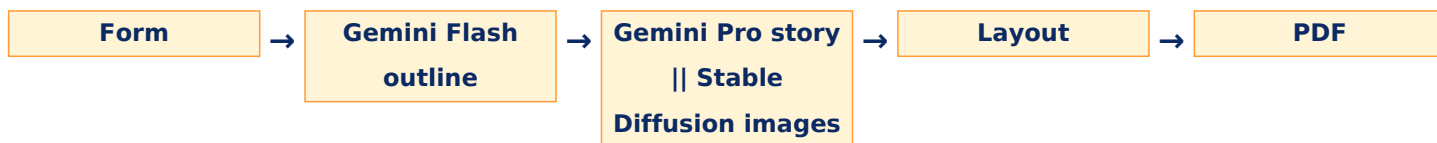
2 Problem and objectives

People without drawing skills cannot easily make comics. ComicCraft generates outline, story, images and layout automatically.

3 Requirements

- Form input: prompt, character, setting, tone, style
- Outline, story, images, preview, PDF
- JSON API and image test endpoint

4 Architecture



5 Implementation

FastAPI routes orchestrate `generate_outline`, `generate_story`, `generate_image`, `build_comic_layout` and `save_pdf`. Story and images are generated in parallel. The image pipeline is cached and warmed up at start.

6 Usage

- Start the server and open the home page
- Enter prompt and character; pick setting, tone, style
- Click Generate Comic and wait
- Review the preview; download the PDF

7 Testing

Smoke tests confirmed outline, story, images and PDF. About 12.4 s per image on Apple M5. See Performance Testing.

8 Limitations

- Gemini free quota: 20 requests per day per model
- No character consistency between panels
- CJK text unsupported in PDF
- Deprecated google-generativeai SDK

Screenshots



Home page

Panel 1: The Threshold of Adventure



Rio, a small orange fox wearing an oversized green scarf and steampunk-style goggles, stands confidently at the entrance of the Whispering Woods. The trees are tall with shimmering silver leaves, and a path of soft moss leads into the depths of the vibrant greenery.

Caption: The threshold of the Whispering Woods awaits.

Rio adjusted his goggles and wrapped his oversized green scarf tighter against the morning chill. "Today is the day," he chirped to the silver trees, his tail twitching with anticipation. He took a bold step forward, ready to uncover the secrets hidden within the glowing canopy.

Exported PDF page

9 Team

Member	Role
Fathima Fahmiya S	Project Lead & Backend Developer
Nivriti Muthuvairan	AI & Image Engineer
Mounika M	Frontend & UI/UX Developer
Tharun P	Testing & QA Engineer

Yuvaraj B	Documentation & Demo Coordinator
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10 Future work

Speech bubbles, character consistency, hosted GPU, SDK migration, CI for the test suite and Docker. See Scalability & Future Plan.